

Monochromatic Union-Closed Families in the Boolean Lattice

By DEEP BHATTACHARJEE[†], PRIYABRATA MANDAL[‡], and USHASHI BHATTACHARYA

Abstract

Let $f(n)$ be the largest integer such that every 2-colouring of the subsets of $\{1, \dots, n\}$ contains a monochromatic family of at least $f(n)$ sets closed under union and intersection, and let $F(n)$ be the analogous quantity for families closed under union only. Erdős and Ulam asked for the order of magnitude of both functions; Erdős recorded the trivial bound $f(n) \geq (n+1)/2$, wrote that there was no plausible conjecture for the true order of $f(n)$, and conjectured that $F(n) > n^c$ for every constant c and that $F(n) < (1+\varepsilon)^n$ for every $\varepsilon > 0$. Random colourings give $f(n) \leq n^2 + n + 1$ and $f(n) < 12n(\log_2(n+2))^2$, and, through the Sauer–Shelah lemma, $F(n) \leq \sum_{j < d} \binom{n}{j}$ whenever $2^d > nd + 2$, so that $F(n) \leq n^{(1+o(1))\log_2 n}$; this proves the second conjecture. The limits of the random method are located: a random colouring contains, in expectation, a monochromatic union-closed family of size $n^{(1+o(1))\log_2 \log_2 n}$ and, with probability at least $1/2$, a monochromatic chain of n sets. Exact values are computed by a counterexample-guided satisfiability search: $f(n) = \lceil (n+1)/2 \rceil$ for $1 \leq n \leq 10$, so that the trivial lower bound is sharp in this range, and $F(n)$ is determined for $n \leq 7$; for $n \leq 10$ the extremal colourings for f can be chosen to depend only on the cardinality of the set, and these colourings are listed. The upper bound for $F(n)$, the resolution of the second conjecture, and the trivial lower bound are formally verified in Lean 4 with Mathlib.

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[†]Corresponding author. [‡]Deep Bhattacharjee and Priyabrata Mandal contributed equally to this work and are joint first authors.

1. Introduction

Write $[n] = \{1, \dots, n\}$ and $2^{[n]}$ for the family of all subsets of $[n]$. A 2-colouring is a map $\chi : 2^{[n]} \rightarrow \{0, 1\}$, and a family $\mathcal{F} \subseteq 2^{[n]}$ is *monochromatic* if χ is constant on $\mathcal{F}$. A family is *union-closed* if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$, and it is a *sublattice* if it is closed under both union and intersection. Every nonempty family with one member is closed under both operations, so the following quantities are well defined:

$$f(n) = \min_{\chi} \max\{|\mathcal{L}| : \mathcal{L} \subseteq 2^{[n]} \text{ a monochromatic sublattice}\},$$

$$F(n) = \min_{\chi} \max\{|\mathcal{F}| : \mathcal{F} \subseteq 2^{[n]} \text{ a monochromatic union-closed family}\},$$

the minima being over all 2-colourings of $2^{[n]}$. Clearly $F(n) \geq f(n)$. A maximal chain $\emptyset \subset \{1\} \subset \{1, 2\} \subset \dots \subset [n]$ has $n + 1$ members, at least $\lceil (n + 1)/2 \rceil$ of one colour, and every subchain is a sublattice; hence

$$F(n) \geq f(n) \geq \left\lceil \frac{n + 1}{2} \right\rceil. \quad (1.1)$$

The problem of estimating $f(n)$ and $F(n)$ was raised by Erdős and Ulam; Erdős recorded it in [Er78], where he wrote that (1.1) was all they could say about $f(n)$ and that they had no plausible conjecture for its true order of magnitude. For $F(n)$ he conjectured that $F(n) > n^c$ for every constant c and all $n > n_0(c)$, and that $F(n) < (1 + \varepsilon)^n$ for every $\varepsilon > 0$ and all large n ; he also recorded, without reference, a result of Howorka that the first conjecture holds for colourings in which all sets of the same cardinality receive the same colour. The problem is Problem 1183 in Bloom's database [Bl26], where it is listed as open with no further progress.

1.1. Results. The first two results are upper bounds obtained from random colourings. Logarithms are to base 2.

THEOREM 1.1. *For every $n \geq 1$,*

$$f(n) \leq n^2 + n + 1 \quad \text{and} \quad f(n) < 12n(\log_2(n + 2))^2.$$

THEOREM 1.2. *Let $n \geq 1$ and let $d \geq 1$ be an integer with $2^d > nd + 2$. Then*

$$F(n) \leq \sum_{j=0}^{d-1} \binom{n}{j}.$$

In particular, for $n \geq 32$ one may take $d = \lceil \log_2 n + \log_2 \log_2 n \rceil + 1$, and then

$$F(n) \leq n^{\log_2 n + \log_2 \log_2 n + 1} = \exp((1 + o(1))(\log_2 n)(\ln n)),$$

so that $F(n)^{1/n} \rightarrow 1$: the second conjecture of Erdős holds, in the strong form $F(n) \leq \exp(O((\log n)^2))$.

Theorem 1.2 together with (1.1) places $F(n)$ between $(n+1)/2$ and $n^{(1+o(1))\log_2 n}$. The next result shows that the random colouring, which is the only tool in Theorems 1.1 and 1.2, cannot close this gap to a polynomial, and cannot bring $f(n)$ below n .

THEOREM 1.3. *Let χ be a uniformly random 2-colouring of $2^{[n]}$.*

- (i) *With probability at least $1/2$ there is a monochromatic chain of at least n sets; consequently, with probability at least $1/2$, there is a monochromatic sublattice of size at least n .*
- (ii) *The expected size of the largest monochromatic union-closed family is at least*

$$\sum_{j=0}^n \binom{n}{j} 2^{1-2^j} \geq \frac{2}{n} \left(\frac{n}{j_0}\right)^{j_0} \quad (n \geq 4, j_0 = \lfloor \log_2 \log_2 n \rfloor),$$

and hence at least $n^{\log_2 \log_2 n - 2 - o(1)}$.

The exponent $\log_2 \log_2 n$ in (ii) and the exponent $\log_2 n$ in Theorem 1.2 are the two sides of the same phenomenon: the union-closed families that a random colouring cannot avoid are the complements of the down-sets $\{Y : \chi([n] \setminus Y') = \chi([n]) \text{ for all } Y' \subseteq Y\}$, which have Vapnik–Chervonenkis dimension about $\log_2 \log_2 n$, while the Sauer–Shelah bound used in Theorem 1.2 only controls the dimension up to about $\log_2 n$.

The last result is computational and concerns small n .

THEOREM 1.4. *The exact values of $f(n)$ and $F(n)$ for small n are as follows.*

n	1	2	3	4	5	6	7	8	9	10
$f(n)$	1	2	2	3	3	4	4	5	5	6
$F(n)$	1	2	2	3	4	5	7			

In particular $f(n) = \lceil (n+1)/2 \rceil$ for $1 \leq n \leq 10$, so that the trivial bound (1.1) is sharp for f in this range. For each $n \leq 10$ there is a colouring attaining $f(n)$ that depends only on the cardinality of the set; such colourings are listed in Section 8.

The computations use a counterexample-guided satisfiability search, described in Section 8, and the program is supplied with the paper.

1.2. *Organisation.* Section 2 collects the structure theory of sublattices of $2^{[n]}$ (join-irreducibles, Birkhoff’s correspondence with down-sets, Dilworth’s theorem) and of union-closed families (independent systems, Vapnik–Chervonenkis dimension, the Sauer–Shelah lemma). Section 3 proves Theorem 1.1, Section 4 proves Theorem 1.2, Section 6 proves Theorem 1.3, Section 8 describes the computations behind Theorem 1.4, Section 10 lists the questions that remain, and Section 11 describes the Lean formalisation.

1.3. *Related work.* The problem studied here is easy to confuse with a more famous one that it resembles only superficially, so the distinction is worth making precise. *Frankl’s union-closed sets conjecture* [Fr79], posed in 1979, concerns a single union-closed family $\mathcal{F} \neq \{\emptyset\}$ of finite sets and asks whether some element lies in at least half the members of $\mathcal{F}$; it is a statement about the internal structure of one family, with no colouring anywhere in it. It was open in general until Gilmer [Gi22] proved in 2022, by an entropy argument, that some element lies in at least a fixed positive proportion (he obtained 0.01) of the members of every union-closed family; the constant was improved to 0.38234 by Alweiss, Huang and Sellke [AHS24] shortly afterwards, with 0.5, the full conjecture, still open at the time of writing. The problem of the present paper is a Ramsey-type question about *every* 2-colouring of the entire Boolean lattice $2^{[n]}$: it asks not whether a given union-closed family is internally balanced, but how large a monochromatic union-closed (or intersection-and-union-closed) family a colouring is forced to contain. The two problems share only their two words “union-closed”; a positive answer to one has no direct bearing on the other, and the entropy method of [Gi22, AHS24], which compares a family to its own down-set, does not seem to have an analogue here, since there is no single family to compare a colour class to before the colour class itself is fixed.

The Sauer–Shelah lemma (Lemma 2.5(c)) originates in extremal set theory [Sa72, Sh72] and is the combinatorial core of the theory of Vapnik–Chervonenkis dimension [VC71] in statistical learning theory and of NIP model theory; its use here, bounding a union-closed family by the independent systems it contains, is close to its original combinatorial form rather than to its statistical applications, and the textbook treatments in [Bo86, Chapter 17] and [St12, Section 3.4] cover both the lemma and Dilworth’s theorem [Di50] used in Lemma 2.2. Erdős’s own problem list [Er78] groups the question of this paper with several other

coloured extremal set-theory problems from the same period; Bloom's continuously updated database of Erdős's problems [B126] records the second conjecture resolved in Theorem 1.2 and the first, left open here, as Problem 1183, and is the source of the precise attributions used throughout.

2. Preliminaries

2.1. *Sublattices of the Boolean lattice.* Throughout this subsection $\mathcal{L} \subseteq 2^{[n]}$ is a sublattice with $|\mathcal{L}| \geq 2$, that is, a nonempty family closed under union and intersection. Put

$$B_0 = \bigcap_{A \in \mathcal{L}} A, \quad T_0 = \bigcup_{A \in \mathcal{L}} A.$$

A member $J \in \mathcal{L}$ is *join-irreducible* if $J \neq B_0$ and J is not the union of the members of $\mathcal{L}$ properly contained in J . Let $\mathcal{J}(\mathcal{L})$ be the set of join-irreducible members and $k = |\mathcal{J}(\mathcal{L})|$. For $x \in T_0 \setminus B_0$ put

$$A_x = \bigcap \{A \in \mathcal{L} : x \in A\} \in \mathcal{L},$$

the smallest member of $\mathcal{L}$ containing x . A *down-set* of a finite poset P is a subset D with $y \leq x \in D \Rightarrow y \in D$; $\text{Down}(P)$ denotes the set of down-sets. The *width* $w(P)$ is the largest size of an antichain.

LEMMA 2.1 (Structure of sublattices). *Let $\mathcal{L}$ be a sublattice of $2^{[n]}$ with $|\mathcal{L}| \geq 2$.*

- (a) *For $x \in T_0 \setminus B_0$, A_x is join-irreducible, and every join-irreducible member is of the form A_x .*
- (b) *Every $A \in \mathcal{L}$ equals $B_0 \cup \bigcup \{J \in \mathcal{J}(\mathcal{L}) : J \subseteq A\}$.*
- (c) *If $J \in \mathcal{J}(\mathcal{L})$ and $J \subseteq A \cup A'$ with $A, A' \in \mathcal{L}$, then $J \subseteq A$ or $J \subseteq A'$. The same holds for unions of any number of members.*
- (d) *The map $A \mapsto D(A) = \{J \in \mathcal{J}(\mathcal{L}) : J \subseteq A\}$ is a bijection from $\mathcal{L}$ onto $\text{Down}(\mathcal{J}(\mathcal{L}), \subseteq)$; in particular $|\mathcal{L}| = |\text{Down}(\mathcal{J}(\mathcal{L}), \subseteq)|$.*
- (e) *$k = |\mathcal{J}(\mathcal{L})| \leq n$, and $w(\mathcal{J}(\mathcal{L}), \subseteq) \leq \log_2 |\mathcal{L}|$.*
- (f) *Define $\psi : [n] \rightarrow \mathcal{J}(\mathcal{L}) \cup \{\perp, \top\}$ by $\psi(x) = \perp$ if $x \in B_0$, $\psi(x) = \top$ if $x \notin T_0$, and $\psi(x) = A_x$ otherwise. Then $\mathcal{L} = \{A_D : D \in \text{Down}(\mathcal{J}(\mathcal{L}), \subseteq)\}$ with $A_D = \{x : \psi(x) = \perp \text{ or } \psi(x) \in D\}$. Consequently $\mathcal{L}$ is determined by the poset $(\mathcal{J}(\mathcal{L}), \subseteq)$, given abstractly on a set of k labels, together with the map ψ .*

Proof. (a) $A_x \in \mathcal{L}$ since $\mathcal{L}$ is closed under intersection and T_0 is a member containing x ; $A_x \neq B_0$ since $x \notin B_0$; and if A_x were the union of members properly contained in it, one of them would contain x , contradicting the minimality of A_x . Conversely let $J \in \mathcal{J}(\mathcal{L})$ and let U be the union of the members of $\mathcal{L}$ properly contained in J ; since $B_0 \subseteq J$ and $B_0 \neq J$, $U \in \mathcal{L}$ and $B_0 \subseteq U \subsetneq J$. Pick $x \in J \setminus U$. Then $A_x \subseteq J$, and $A_x \subsetneq J$ would give $A_x \subseteq U$, which is false; so $A_x = J$.

(b) Let $A \in \mathcal{L}$. Then $B_0 \subseteq A$, and for $x \in A \setminus B_0$ we have $A_x \subseteq A$ with $A_x \in \mathcal{J}(\mathcal{L})$ by (a), so A is contained in the right side; the reverse inclusion is clear.

(c) Since $\mathcal{L}$ is closed under intersection, $J = J \cap (A \cup A') = (J \cap A) \cup (J \cap A')$ with $J \cap A, J \cap A' \in \mathcal{L}$ contained in J . If both were proper subsets of J , then J would be the union of two members properly contained in it, contradicting join-irreducibility. Hence $J \cap A = J$ or $J \cap A' = J$. For a union of several members apply this repeatedly.

(d) By (b) the map is injective. It takes values in down-sets: if $J' \subseteq J \subseteq A$ then $J' \subseteq A$. For surjectivity let D be a down-set and put $A_D = B_0 \cup \bigcup_{J \in D} J \in \mathcal{L}$. If $J \in \mathcal{J}(\mathcal{L})$ satisfies $J \subseteq A_D$, then by (c) applied to the union $B_0 \cup \bigcup_{J' \in D} J'$ we have $J \subseteq B_0$, impossible as $B_0 \subsetneq J$, or $J \subseteq J'$ for some $J' \in D$, whence $J \in D$ as D is a down-set. Thus $D(A_D) = D$.

(e) Order $\mathcal{J}(\mathcal{L})$ by a linear extension $J_1, \dots, J_k$ of $\subseteq$; the sets $D_i = \{J_1, \dots, J_i\}$, $0 \leq i \leq k$, are down-sets, so by (d) $A_{D_0} \subsetneq A_{D_1} \subsetneq \dots \subsetneq A_{D_k}$ is a chain of $k+1$ distinct subsets of $[n]$, whence $k+1 \leq n+1$. If S is an antichain of $\mathcal{J}(\mathcal{L})$, the $2^{|S|}$ sets $\{J \in \mathcal{J}(\mathcal{L}) : J \subseteq J' \text{ for some } J' \in S'\}$, $S' \subseteq S$, are distinct down-sets, so $2^w \leq |\mathcal{L}|$.

(f) By (d) every member is $A_D = B_0 \cup \bigcup_{J \in D} J$ for a unique down-set D . An element $x \in [n]$ lies in A_D if and only if $x \in B_0$, or $x \in T_0 \setminus B_0$ and $x \in J$ for some $J \in D$; and for $x \in T_0 \setminus B_0$, $x \in J$ holds if and only if $A_x \subseteq J$, by minimality of A_x . Since D is a down-set, $A_x \subseteq J \in D$ for some J if and only if $A_x \in D$. This is the displayed description of A_D , which uses only ψ and the order structure of $\mathcal{J}(\mathcal{L})$. $\square$

Part (d) is Birkhoff's representation of finite distributive lattices [St12, Theorem 3.4.1], specialised to sublattices of $2^{[n]}$; we have included the short proof since (c) and (f) are used in exactly this form.

LEMMA 2.2 (Counting posets of bounded width). *The number of partial orders on the labelled set $[k]$ of width at most w is at most $(k+1)^{k(w+2)}$.*

Proof. Let P be such a poset. By Dilworth's theorem [Di50], $[k]$ is the disjoint union of w chains $C_1, \dots, C_w$ of P , some possibly empty. Record the following data: for each $x \in [k]$ the index $c(x) \in [w]$ of its chain and its position $p(x) \in [k]$ within that chain (counted from the bottom); and for each $x \in [k]$ and each $i \in [w]$ the number $r_i(x) \in \{0, \dots, k\}$ of elements $y \in C_i$ with $y \leq x$. Since $y \leq x$ and $y' \leq y$ with $y' \in C_i$ imply $y' \leq x$, the elements of C_i below x form an initial segment of C_i , and so $y \leq x$ holds if and only if $p(y) \leq r_{c(y)}(x)$. Thus the data determine P , and the number of possible data is at most $w^k \cdot k^k \cdot (k+1)^{kw} \leq (k+1)^{k(w+2)}$, using $w \leq k$. $\square$

LEMMA 2.3 (Counting sublattices). *For $m \geq 2$, the number $N(m)$ of sublattices $\mathcal{L} \subseteq 2^{[n]}$ with $|\mathcal{L}| = m$ satisfies*

$$N(m) \leq \sum_{k=1}^n (k+1)^{k(\lceil \log_2 m \rceil + 2)} (k+2)^n \leq (n+2)^{n(\log_2 m + 4)}.$$

Moreover the total number of sublattices of $2^{[n]}$ with at least two members is at most $2^{n(n+1)}$.

Proof. By Lemma 2.1(f), $\mathcal{L}$ is determined by a poset on k labels together with a map $\psi : [n] \rightarrow [k] \cup \{\perp, \top\}$, where $k \leq n$ and the poset has width at most $\log_2 m$ by Lemma 2.1(e). Lemma 2.2 bounds the number of posets and $(k+2)^n$ bounds the number of maps, which gives the first inequality. For the second, each summand is at most $(n+1)^{n(\log_2 m + 2)} (n+2)^n \leq (n+2)^{n(\log_2 m + 2) + n}$, and there are $n \leq (n+2)^n$ summands. For the last assertion, $\mathcal{L}$ is determined by B_0 together with $\mathcal{J}(\mathcal{L})$, by Lemma 2.1(b) and the fact that $\mathcal{L}$ consists of all unions $B_0 \cup \bigcup_{J \in D} J$; this is a list of at most $n+1$ subsets of $[n]$, and there are at most $(2^n)^{n+1}$ such lists. $\square$

2.2. Union-closed families.

Definition 2.4. Sets $A_1, \dots, A_d \subseteq [n]$ form an *independent system* if each A_i has a *private element*: an $x_i \in A_i$ with $x_i \notin A_j$ for all $j \neq i$. The *independence number* $\text{ind}(\mathcal{F})$ of a family $\mathcal{F}$ is the largest d for which $\mathcal{F}$ contains an independent system of d sets. A set $X \subseteq [n]$ is *shattered* by $\mathcal{F}$ if $\{A \cap X : A \in \mathcal{F}\} = 2^X$, and the *Vapnik–Chervonenkis dimension* $\text{vc}(\mathcal{F})$ is the largest size of a shattered set.

- LEMMA 2.5. (a) *If $A_1, \dots, A_d$ is an independent system, then the $2^d - 1$ unions $\bigcup_{i \in I} A_i$, $\emptyset \neq I \subseteq [d]$, are pairwise distinct.*
 (b) *For every family $\mathcal{F}$, $\text{vc}(\mathcal{F}) \leq \text{ind}(\mathcal{F})$.*

(c) (Sauer [Sa72], Shelah [Sh72].) If $\text{vc}(\mathcal{F}) \leq d$ then $|\mathcal{F}| \leq \sum_{j=0}^d \binom{n}{j}$.

Proof. (a) The private element x_i lies in $\bigcup_{i \in I} A_i$ if and only if $i \in I$, so the union determines I . (b) Let $X = \{x_1, \dots, x_d\}$ be shattered, and for each i choose $A_i \in \mathcal{F}$ with $A_i \cap X = \{x_i\}$. Then x_i is a private element of A_i in the system $A_1, \dots, A_d$. (c) This is the Sauer–Shelah lemma. $\square$

3. Upper bounds for $f(n)$

Let χ be a uniformly random 2-colouring of $2^{[n]}$: the colours $\chi(A)$, $A \subseteq [n]$, are independent and uniform on $\{0, 1\}$. A fixed family of m sets is monochromatic with probability 2^{1-m} .

Proof of Theorem 1.1, first bound. Let $M = n^2 + n + 2$. By Lemma 2.3 there are at most $2^{n(n+1)}$ sublattices with at least two members, so the expected number of monochromatic sublattices with at least M members is at most $2^{n(n+1)} \cdot 2^{1-M} = 2^{-1} < 1$. Hence some colouring has no monochromatic sublattice with M or more members, and $f(n) \leq M - 1$. $\square$

Proof of Theorem 1.1, second bound. For $n \leq 2$ the bound is weaker than the trivial $f(n) \leq 2^n$, so let $n \geq 3$ and put $\ell = \log_2(n+2) \geq \log_2 5 > 2.32$, $A = n\ell$ and $M = \lceil 12n\ell^2 \rceil$. By Lemma 2.3, for every $m \geq 2$,

$$N(m) 2^{1-m} \leq 2^{A(\log_2 m + 4) + 1 - m}.$$

Let $\varphi(m) = m/2 - A(\log_2 m + 4) - 1$. We claim $\varphi(m) \geq 0$ for all real $m \geq M$. The derivative $\varphi'(m) = 1/2 - A/(\ln 2)$ is positive for $m > 2A/\ln 2$, and $M \geq 12n\ell^2 > 2.9n\ell > 2A/\ln 2$, so it suffices to check $\varphi(M) \geq 0$. Using $M \leq 13n\ell^2$ and $\log_2 \ell \leq \ell - 1$ (valid for $\ell \geq 1$),

$$\log_2 M + 4 \leq \log_2 13 + \log_2 n + 2\log_2 \ell + 4 \leq 3.71 + \ell + 2(\ell - 1) + 4 = 3\ell + 5.71,$$

so $A(\log_2 M + 4) + 1 \leq 3n\ell^2 + 5.71n\ell + 1 \leq 6n\ell^2 \leq M/2$, the middle inequality being $3\ell \geq 5.71 + 1/(n\ell)$, which holds as $\ell > 2.32$. Hence, for $m \geq M$, $N(m)2^{1-m} \leq 2^{-m/2}$, and the expected number of monochromatic sublattices with at least M members is at most $\sum_{m \geq M} 2^{-m/2} < 2^{-M/2}/(1 - 2^{-1/2}) < 1$. Some colouring therefore has no monochromatic sublattice with M or more members, and $f(n) \leq M - 1 < 12n\ell^2$. $\square$

Remark 3.1. The first bound is the better one for $n \leq 1500$ or so, the second thereafter. Both use only the number of sublattices and the probability 2^{1-m} ; the second is limited by the factor $(n+2)^n$ in Lemma 2.3, which is really present (there are $n!$ maximal chains), so this method cannot give

$f(n) = O(n)$. Theorem 1.3(i) shows that the random colouring itself has monochromatic chains of n sets, so no analysis of the random colouring can bring the bound below n .

4. Upper bound for $F(n)$

Proof of Theorem 1.2. Let χ be a uniformly random 2-colouring and let $d \geq 1$ satisfy $2^d > nd + 2$. Call an ordered d -tuple $(A_1, \dots, A_d)$ of subsets of $[n]$ *bad* if it is an independent system and all $2^d - 1$ unions $\bigcup_{i \in I} A_i$, $\emptyset \neq I \subseteq [d]$, have the same colour. By Lemma 2.5(a) these unions are distinct sets, so a fixed independent tuple is bad with probability $2^{1-(2^d-1)} = 2^{2-2^d}$. There are at most 2^{nd} tuples, so the expected number of bad tuples is at most $2^{nd+2-2^d} < 1$. Fix a colouring with no bad tuple.

Let $\mathcal{F}$ be a monochromatic union-closed family for this colouring. If $\mathcal{F}$ contained an independent system $A_1, \dots, A_d$, then all unions $\bigcup_{i \in I} A_i$ would belong to $\mathcal{F}$ by union-closure and would therefore be monochromatic, making the tuple bad. Hence $\text{ind}(\mathcal{F}) \leq d - 1$, so $\text{vc}(\mathcal{F}) \leq d - 1$ by Lemma 2.5(b), and $|\mathcal{F}| \leq \sum_{j \leq d-1} \binom{n}{j}$ by Lemma 2.5(c). This proves the first assertion.

For the second, let $n \geq 32$ and $d = \lceil \log_2 n + \log_2 \log_2 n \rceil + 1$. Then $2^d \geq 2n \log_2 n$, while $nd + 2 \leq n(\log_2 n + \log_2 \log_2 n + 2) + 2$; the inequality $2n \log_2 n > n \log_2 n + n \log_2 \log_2 n + 2n + 2$ is equivalent to $\log_2 n - \log_2 \log_2 n > 2 + 2/n$, which holds for $n \geq 32$ (at $n = 32$ the left side is $5 - \log_2 5 > 2.67$ and the left side is increasing). Finally $\sum_{j \leq d-1} \binom{n}{j} \leq n^{d-1} \leq n^{\log_2 n + \log_2 \log_2 n + 1}$, and $n^{\log_2 n + \log_2 \log_2 n + 1} = \exp((\log_2 n + \log_2 \log_2 n + 1) \ln n)$, whose n -th root tends to 1. $\square$

Remark 4.1. For $n < 32$ the theorem applies with other values of d : $d = 6$ for $n \leq 10$, $d = 7$ for $n \leq 18$, $d = 8$ for $n \leq 31$. The resulting bounds are far from the true values in Theorem 1.4, as they must be, since the method does not see the chain structure at all.

Remark 4.2 (An explicit correction). The inequality $\sum_{j \leq d-1} \binom{n}{j} \leq n^{d-1}$ used at the end of the proof of Theorem 1.2 is not quite literally true: since $\binom{n}{j} \leq n^j$ for every $j \geq 0$ and $n \geq 1$, one has only

$$\sum_{j=0}^{d-1} \binom{n}{j} \leq \sum_{j=0}^{d-1} n^j \leq d n^{d-1} \quad (n \geq 1),$$

the last step bounding each of the d terms by the largest, n^{d-1} . This is the correct form of the estimate, and it changes nothing: with $d = \lceil \log_2 n +$

$\log_2 \log_2 n \rfloor + 1 = O(\log n)$, the extra factor d contributes $\log_2 d = O(\log \log n)$ to $\log_2$ of the bound, which is already absorbed by the $o(1)$ in the exponent $\log_2 n + \log_2 \log_2 n + 1$. Precisely,

$$F(n) \leq d \cdot n^{d-1} \leq (\log_2 n + \log_2 \log_2 n + 2) \cdot n^{\log_2 n + \log_2 \log_2 n + 1} \quad (n \geq 32), \quad (4.1)$$

which is the fully justified version of the last display in the proof of Theorem 1.2 and is the bound tabulated in Table 1 below.

5. Explicit and numerical bounds

Theorem 1.2 is stated for $n \geq 32$ so that the two-line computation in its proof goes through without case analysis; the underlying first-moment argument needs no such threshold, and the following restatement removes it.

PROPOSITION 5.1 (Threshold-free form). *For every $n \geq 1$, let $d(n)$ be the least positive integer d with $2^d > nd + 2$. Then*

$$F(n) \leq \sum_{j=0}^{d(n)-1} \binom{n}{j} \leq d(n) n^{d(n)-1}.$$

The function $d(n)$ is nondecreasing, takes the values 3, 4, 4, 5, 5, 6, 6, 6, 6, 6 for $n = 1, \dots, 10$, and satisfies $d(n) = \log_2 n + \log_2 \log_2 n + O(1)$ as $n \rightarrow \infty$; in particular $d(n)/\log_2 n \rightarrow 1$, so $F(n) \leq n^{(1+o(1))\log_2 n}$ and $F(n)^{1/n} \rightarrow 1$ with no restriction on n .

Proof. The first inequality is the first assertion of Theorem 1.2, proved for every $n \geq 1$ and every d with $2^d > nd + 2$ (the proof uses only this hypothesis, not $n \geq 32$); taking $d = d(n)$, the least such d , gives the strongest instance. The second inequality is Remark 4.2. Monotonicity of $d(n)$ holds because $2^d > nd + 2$ implies $2^d > n'd + 2$ for $n' \leq n$. The values for $n \leq 9$ and the asymptotics for $d(n)$ are immediate from the defining inequality: writing $d = \log_2 n + \log_2 \log_2 n + e(n)$, the condition $2^d > nd + 2$ becomes, after dividing by $n \log_2 n$, an inequality in $e(n)$ that is satisfied once $e(n) \geq 2 + o(1)$ and fails once $e(n) \leq -o(1)$, giving $e(n) = O(1)$; this is the computation carried out for $n \geq 32$ in the proof of Theorem 1.2 and it is valid, with different explicit constants, for every n . $\square$

Table 1 evaluates the bound of Proposition 5.1 directly: for each n it records $d(n)$, the exact value of $\log_2 \sum_{j < d(n)} \binom{n}{j}$ for $n \leq 2000$ (computed by exact integer arithmetic) or the bound $\log_2(d(n)) + (d(n) - 1) \log_2 n$ of (4.1) for

larger n , and the ratio of this quantity to n , which is $\log_2(F(n)^{1/n})$ up to the direction of the inequality and which Theorem 1.2 and Proposition 5.1 assert tends to 0.

n	$d(n)$	$\log_2(\text{bound})$	$\log_2(\text{bound})/n$
10	6	9.317	0.9317
32	9	23.842	0.7451
50	9	29.287	0.5857
100	10	40.937	0.4094
200	12	58.514	0.2926
500	13	78.598	0.1572
1000	14	96.925	0.0969
10^4	18	225.891	0.02259
10^5	22	348.802	0.00349
10^6	25	478.358	0.00048

Table 1. The bound of Proposition 5.1; entries for $n \leq 2000$ use the exact value of $\log_2 \sum_{j < d(n)} \binom{n}{j}$, entries for $n > 2000$ use the right side of (4.1). The last column, which is at least $\log_2(F(n))/n$ by Theorem 1.2 and at most the value shown, illustrates the convergence $F(n)^{1/n} \rightarrow 1$ at concrete scales; it says nothing about the true order of $F(n)$, which by Theorem 1.3(ii) is not polynomial along the random colouring and is addressed only for cardinality colourings, in Section 7.

6. Limits of the random colouring

Proof of Theorem 1.3. (i) Build a chain greedily. Put $S_0 = \emptyset$ and $c = \chi(\emptyset)$. Given S_i with $|S_i| = i < n$ and $\chi(S_i) = c$, the $n - i$ sets $S_i \cup \{x\}$, $x \notin S_i$, have independent uniform colours, none of which has been examined before (each is examined only at the step at which it is a candidate, and S_i determines the step). The step fails if all of them have colour $1 - c$, which has probability $2^{-(n-i)}$; otherwise choose one of colour c as S_{i+1} . The probability that no step fails for $i = 0, \dots, n - 2$ is at least $1 - \sum_{i=0}^{n-2} 2^{-(n-i)} = 1 - \sum_{j=2}^n 2^{-j} > 1/2$, and in that event $S_0 \subset S_1 \subset \dots \subset S_{n-1}$ is a monochromatic chain of n sets, which is a sublattice.

(ii) Let $T = [n]$ and $c = \chi(T)$, and let

$$\mathcal{Y} = \{Y \subseteq T : \chi(T \setminus Y') = c \text{ for all } Y' \subseteq Y\}.$$

Then $\mathcal{Y}$ is a down-set of $2^{[n]}$ containing $\emptyset$, and the family $\mathcal{F} = \{T \setminus Y : Y \in \mathcal{Y}\}$ is monochromatic of colour c and union-closed: $(T \setminus Y) \cup (T \setminus Y') = T \setminus (Y \cap Y')$ and $Y \cap Y' \in \mathcal{Y}$. A set Y with $|Y| = j$ lies in $\mathcal{Y}$ if and only if the $2^j - 1$ sets $T \setminus Y'$, $\emptyset \neq Y' \subseteq Y$, all have the colour of T , an event of probability $2^{-(2^j-1)}$. Hence $\mathbb{E}|\mathcal{F}| = \mathbb{E}|\mathcal{Y}| = \sum_{j=0}^n \binom{n}{j} 2^{1-2^j}$, and the largest monochromatic union-closed family has expected size at least this. For the numerical bound take $j_0 = \lfloor \log_2 \log_2 n \rfloor$ with $n \geq 4$; then $2^{j_0} \leq \log_2 n$, so $2^{1-2^{j_0}} \geq 2/n$, and $\binom{n}{j_0} \geq (n/j_0)^{j_0}$; thus $\mathbb{E}|\mathcal{F}| \geq 2(n/j_0)^{j_0}/n$. As $(n/j_0)^{j_0} = n^{j_0-o(1)}$ and $j_0 \geq \log_2 \log_2 n - 1$, this is $n^{\log_2 \log_2 n - 2-o(1)}$. $\square$

Remark 6.1. Part (ii) shows that no bound of the form $F(n) \leq n^C$ with a constant C can be proved by exhibiting a random colouring; the same family also shows that for the random colouring the Vapnik–Chervonenkis dimension of the largest monochromatic union-closed family is at least about $\log_2 \log_2 n$, since $\mathcal{Y}$ contains all sets of size at most j_0 with positive probability bounded away from zero when $2^{j_0} \leq \log_2 n$.

7. Colourings depending only on cardinality

A colouring χ is a *cardinality colouring* if $\chi(S) = g(|S|)$ for some $g : \{0, 1, \dots, n\} \rightarrow \{0, 1\}$. These are the colourings for which Erdős recorded Howorka’s assertion, and they are the colourings that attain $f(n)$ for $n \leq 10$ (Section 8). For a cardinality colouring, monochromatic closed families can be built from monochromatic arithmetic progressions of g .

PROPOSITION 7.1. *Let $\chi(S) = g(|S|)$ and suppose that the arithmetic progression $a, a + s, \dots, a + (k-1)s$, with $s \geq 1$, $k \geq 1$ and $a + (k-1)s \leq n$, is monochromatic for g . Put $q = \lfloor a/s \rfloor$ and $b = q + k - 1$. Then there is a monochromatic union-closed family with $\sum_{i=0}^{k-1} \binom{b}{i}$ members, and a monochromatic sublattice with 2^{k-1} members. More generally, if $t_1 < t_2 < \dots < t_L$ and $s, w \geq 1$ are such that $t_L + ws \leq n$ and all the numbers $t_j + is$ ($1 \leq j \leq L$, $0 \leq i \leq w$) have the same colour under g , then there is a monochromatic sublattice with $L 2^w$ members.*

Proof. Put $a_0 = a - qs \in [0, s)$ and choose pairwise disjoint sets $B, X_1, \dots, X_b \subseteq [n]$ with $|B| = a_0$ and $|X_i| = s$; this is possible since $a_0 + bs = a + (k-1)s \leq n$. Let

$$\mathcal{F} = \left\{ B \cup \bigcup_{i \in I} X_i : I \subseteq [b], |I| \geq q \right\}.$$

A member with index set I has cardinality $a_0 + |I|s = a + (|I| - q)s$ with $0 \leq |I| - q \leq k - 1$, so all members have the colour of the progression. The union of the members with index sets I, I' has index set $I \cup I'$, of size at least q , so $\mathcal{F}$ is union-closed, and $|\mathcal{F}| = \sum_{i=q}^b \binom{b}{i} = \sum_{i=0}^{k-1} \binom{b}{i}$. For the sublattice take $|B'| = a$ and $k - 1$ disjoint blocks $X_1, \dots, X_{k-1}$ of size s disjoint from B' , and the family of all $B' \cup \bigcup_{i \in I} X_i$, $I \subseteq [k - 1]$; it is closed under union and intersection, has 2^{k-1} members, and their cardinalities $a + |I|s$ lie in the progression.

For the last assertion choose disjoint blocks $X_1, \dots, X_w$ of size s and a chain $C_1 \subset \dots \subset C_L$ of subsets of $[n] \setminus (X_1 \cup \dots \cup X_w)$ with $|C_j| = t_j$, possible since $t_L \leq n - ws$. The family $\{C_j \cup \bigcup_{i \in I} X_i : 1 \leq j \leq L, I \subseteq [w]\}$ has $L2^w$ members of cardinalities $t_j + |I|s$, all of one colour, and it is closed under union and intersection since $(C_j \cup X_I) \cup (C_{j'} \cup X_{I'}) = C_{\max(j, j')} \cup X_{I \cup I'}$ and $(C_j \cup X_I) \cap (C_{j'} \cup X_{I'}) = C_{\min(j, j')} \cup X_{I \cap I'}$. $\square$

Let $W(k)$ denote the van der Waerden number: the least N such that every 2-colouring of $\{1, \dots, N\}$ contains a monochromatic arithmetic progression of k terms; it is finite by van der Waerden's theorem [vdW27]. The following statement is the one attributed to Howorka in [Er78]; we have not been able to locate a published proof, and give one.

THEOREM 7.2. *Let $k \geq 2$ and $n \geq 2W(k)$. For every cardinality colouring of $2^{[n]}$ there is a monochromatic union-closed family with at least*

$$\binom{\lfloor n/(2W(k)) \rfloor + k - 1}{k - 1} \geq \left(\frac{n}{2(k - 1)W(k)} \right)^{k-1}$$

members, and a monochromatic sublattice with 2^{k-1} members. Consequently, for every constant c , every cardinality colouring of $2^{[n]}$ with n large in terms of c has a monochromatic union-closed family of more than n^c members; and there is a function $\omega(n) \rightarrow \infty$ such that every cardinality colouring has a monochromatic union-closed family of more than $n^{\omega(n)}$ members.

Proof. Let $\chi(S) = g(|S|)$ and $W = W(k)$. The W consecutive integers $m_0, m_0 + 1, \dots, m_0 + W - 1$ with $m_0 = \lceil n/2 \rceil$ lie in $\{0, \dots, n\}$ because $n \geq 2W$, and by the definition of $W(k)$ the colouring g restricted to them contains a monochromatic progression $a, a + s, \dots, a + (k - 1)s$ with $a \geq n/2$ and $(k - 1)s \leq W - 1$, so that $s < W$ and $\lfloor a/s \rfloor \geq \lfloor n/(2W) \rfloor$. By Proposition 7.1 there is a monochromatic union-closed family with $\sum_{i < k} \binom{b}{i} \geq \binom{b}{k-1}$ members, where $b = \lfloor a/s \rfloor + k - 1 \geq \lfloor n/(2W) \rfloor + k - 1$, and $\binom{b}{k-1} \geq (b/(k - 1))^{k-1} \geq$

$(n/(2(k-1)W))^{k-1}$, using $b \geq \lfloor n/(2W) \rfloor + k - 1 \geq n/(2W)$ (as $k \geq 2$). The sublattice statement is the second part of Proposition 7.1. For fixed k the bound is $c_k n^{k-1}$ with $c_k > 0$, which exceeds n^c for n large once $k - 1 > c$. Finally let $\omega(n)$ be the largest k with $2(k-1)W(k) \leq n^{1/2}$; then $\omega(n) \rightarrow \infty$, and the bound is at least $n^{(\omega(n)-1)/2}$, which exceeds $n^{\omega(n)/3}$ for $\omega(n) \geq 3$; replacing ω by $\omega/3$ gives the statement. $\square$

Remark 7.3. The van der Waerden numbers grow so fast that Theorem 7.2 gives nothing for small n ; its content is the polynomial growth for every fixed k . Proposition 7.1 also explains the shape of the extremal cardinality colourings of Section 8: a cardinality colouring with no monochromatic sublattice of more than $\lceil (n+1)/2 \rceil$ members must have no monochromatic progression of k terms with $2^{k-1} > \lceil (n+1)/2 \rceil$, and, for every s and w , at most $\lceil (n+1)/2 \rceil 2^{-w}$ starting points t of one colour with $t, t+s, \dots, t+ws$ all of that colour. For $n \leq 10$ both conditions leave room, as the colourings g_n listed in Section 8 show.

8. Exact values for small n

8.1. *The algorithm.* Fix n and a target M . We wish to decide whether there is a 2-colouring of $2^{[n]}$ with no monochromatic closed family of M or more members, closure meaning union and intersection for f and union alone for F . This is a satisfiability problem in the 2^n Boolean variables x_S , $S \subseteq [n]$, with one clause pair for every closed family $\mathcal{F}$ of size M : not all x_S , $S \in \mathcal{F}$, true, and not all false. Rather than enumerating all closed families in advance, the program proceeds by counterexample-guided refinement: a SAT solver (CaDiCaL, through the PySAT interface [IMM18]) proposes a colouring subject to the clauses collected so far; a branch-and-bound search then looks, in each colour class, for a closed subfamily with at least M members, building families by adding sets one at a time and closing under the operations; if one is found, its clause pair is added and the solver is run again; if none is found, the colouring is a witness that the value is less than M ; if the solver reports unsatisfiability, every colouring has a monochromatic closed family of at least M members and the value is at least M . The symmetry $\chi \mapsto 1 - \chi$ is broken by fixing the colour of $\emptyset$. Increasing M from 1 gives the exact value.

8.2. *Results and checks.* The values in Theorem 1.4 were obtained in this way. The computation for $f(9)$ examined 14 190 clauses and took twenty minutes on one processor core; the computation for $F(7)$ examined 256 428 clauses.

Three independent checks were made. For $n \leq 4$ all 2^{2^n} colourings were enumerated directly and the maximal monochromatic closed family was found by inspecting all closed families, without any SAT solver; the values agree. For every entry of the table the final colouring produced by the solver (the witness that the value is not larger) was verified by a second program that encodes “this colour class contains a closed family of at least M members” as a satisfiability instance with a cardinality constraint, independent of the branch-and-bound search, and the instance was unsatisfiable for both colour classes in every case; the witnesses are supplied with the code, and this second, independent check was re-run for the present version of the paper, confirming for every g_n , $3 \leq n \leq 10$, that neither colour class contains a monochromatic sublattice of $\lceil (n+1)/2 \rceil + 1$ or more members. In particular the values of $f(n)$ for $n \leq 10$ do not depend on any unsatisfiability report: the lower bound is (1.1), and the upper bound is certified by the cardinality colourings g_n listed below, each checked by the second program. The lower bounds for $F(5)$, $F(6)$ and $F(7)$ rest on the solver’s unsatisfiability reports.

The extremal colourings found by the solver for $n \leq 7$ happened to be cardinality colourings; for $n = 8$ and $n = 9$ a cardinality colouring attaining $f(n)$ was found by adding to the solver the clauses $x_S \leftrightarrow x_{S'}$ for $|S| = |S'|$, and for $n = 10$ (where the unrestricted search was not run to completion) by enumerating the cardinality colourings that pass the necessary conditions of Proposition 7.1 and checking each with the second program; since $f(10) \geq 6$ by (1.1), this determines $f(10) = 6$. Writing $g_n = (g_n(0), g_n(1), \dots, g_n(n))$ for the colour of a set of each cardinality, the following colourings have no monochromatic sublattice with more than $\lceil (n+1)/2 \rceil$ members:

$$\begin{aligned} g_3 &= (1, 1, 0, 0), & g_4 &= (1, 1, 0, 1, 0), \\ g_5 &= (1, 1, 0, 1, 0, 0), & g_6 &= (1, 0, 0, 1, 0, 1, 1), \\ g_7 &= (1, 0, 0, 1, 1, 0, 0, 1), & g_8 &= (1, 0, 1, 0, 0, 1, 1, 0, 1), \\ g_9 &= (1, 0, 1, 0, 0, 1, 1, 0, 1, 0), & g_{10} &= (1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1). \end{aligned}$$

For a colouring depending only on cardinality, a monochromatic sublattice corresponds, by Lemma 2.1, to a poset on blocks of elements together with block sizes and a base size, all of whose down-set weights lie in one colour class; the reader may check by hand, for instance, that for g_9 the colour class $\{0, 2, 5, 6, 8\}$ contains the arithmetic progression 2, 5, 8, which yields the sublattice $\{\emptyset, B, B \cup X, B \cup Y, B \cup X \cup Y\}$ of five members with $|B| = 2$, $|X| = |Y| = 3$, and that this family cannot be extended within the colour class.

8.3. *Discussion.* The values show that the trivial bound (1.1) for f is exact up to $n = 10$, and that F exceeds f from $n = 5$ on. We do not conjecture that $f(n) = \lceil (n+1)/2 \rceil$ holds for all n . By Proposition 7.1, a cardinality colouring attaining this value for all n would have to avoid monochromatic arithmetic progressions of length $\log_2(n+1) + 1$ in every colour class, and to contain, for every difference s and length $w + 1$, at most $\lceil (n+1)/2 \rceil 2^{-w}$ monochromatic progressions of that difference and length in each colour; the second condition is met with room to spare for $n \leq 10$ and asks, for large n , for a colouring whose progression counts never exceed their expected values, which we do not expect to exist. Whether $f(n) = O(n)$ is a different question, since general colourings are not constrained by Proposition 7.1.

9. Summary of what is proved and what is open

The results above touch four different quantities, and it is easy to lose track of which are settled and which are not; this section collects them in one place.

The one item in the table that is genuinely settled, Erdős's second conjecture, was open, so far as the sources in Section 1.3 record, from 1978 until the argument of Theorem 1.2; the argument is short (Lemma 2.5 plus a first-moment count) once it is seen that the two-colouring, forced-monochromatic-union structure of the problem is exactly what an independent system detects. The first conjecture is the harder of Erdős's two: even the far more permissive class of cardinality colourings needs the full strength of van der Waerden's theorem (Theorem 7.2) to rule out an n^c bound, and for general colourings no argument in this paper, or, so far as the authors are aware, in the literature, gives any superlinear lower bound on $F(n)$ at all beyond the trivial $\lceil (n+1)/2 \rceil$. The gap between what a random colouring is forced to contain (Theorem 1.3(ii), exponent $\log_2 \log_2 n$) and what the Sauer–Shelah bound permits (Theorem 1.2, exponent $\log_2 n$) is the precise sense in which the first conjecture remains open: closing it either upward, by a better construction than the random colouring, or downward, by a bound sharper than Sauer–Shelah, would resolve Question 2 below, and either direction looks, to the authors, to need an idea not present in this paper.

10. Questions

QUESTION 10.1. *Is $f(n) = \lceil (n+1)/2 \rceil$ for all n ? If not, what is the least n for which $f(n) > \lceil (n+1)/2 \rceil$, and is $f(n) = O(n)$?*

Question	Status	Where
Erdős's second conjecture, $F(n) < (1 + \varepsilon)^n$ for every $\varepsilon > 0$	PROVED, in the strong form $F(n) \leq n^{(1+o(1))\log_2 n}$, for every $n \geq 1$	Theorem 1.2, Prop. 5.1
Erdős's first conjecture, $F(n) > n^c$ for every constant c	OPEN in general; PROVED for cardinality colourings	Question 3, Theorem 7.2
Order of magnitude of $f(n)$	OPEN; $f(n) = \lceil (n+1)/2 \rceil$ for $n \leq 10$, upper bound $f(n) < 12n(\log_2 n)^2$ in general	Theorems 1.1, 1.4
Is $f(n) = O(n)$?	OPEN; the random colouring alone cannot show it, since it already has monochromatic chains of length n	Theorem 1.3(i)
Is $F(n)$ polynomially bounded?	OPEN; known bounds place $\log_2 F(n)$ between about $(\log_2 n)(\log_2 \log_2 n)$ and $2 \log_2 n \cdot \log_2 n$	Theorems 1.2, 1.3(ii)
Exact values of $f(n)$, $F(n)$ for small n	COMPUTED and doubly verified for $n \leq 10$ (partially $F(8)$ – $F(10)$ open)	Theorem 1.4, Appendix A

Table 2. The state of Erdős and Ulam's problem after this paper. Only the second row of the second conjecture is fully closed; every other row records either a gap between a proved upper and a proved lower bound, or an outright open question, and none of the further theorems of this paper depend on any of the open rows.

QUESTION 10.2. *Is $F(n)$ bounded by a polynomial in n ? By Theorem 1.3(ii) a positive answer cannot come from a random colouring; by Theorem 1.2 the true exponent of $\log n$ in $\log F(n)$ is at most 2.*

QUESTION 10.3. *Is $F(n) > n^c$ for every constant c and all large n , as Erdős conjectured? The computed values $F(5) = 4$, $F(6) = 5$, $F(7) = 7$ show that F pulls away from f early, but no lower bound better than $\lceil (n+1)/2 \rceil$ is*

known for general colourings; for cardinality colourings Theorem 7.2 settles the question.

Remark 10.4 (An approach that does not work, and precisely why). A natural strategy for the first conjecture is to import a Ramsey theorem for the Boolean lattice itself. The *poset Ramsey number* $R(Q_k, Q_k)$ is the least n such that every 2-colouring of the elements of $2^{[n]}$ contains a monochromatic copy of the Boolean lattice $Q_k = 2^{[k]}$, where a copy means an order-embedding $\phi : 2^{[k]} \rightarrow 2^{[n]}$, that is, an injection with $A \subseteq B \Leftrightarrow \phi(A) \subseteq \phi(B)$. Axenovich and Walzer [AW17] proved $R(Q_k, Q_k) \geq 2k$, and Lu and Thompson [LT22] proved $R(Q_k, Q_k) \leq k^2 - k + 2$, so this Ramsey number is polynomial in k . If a monochromatic copy of Q_k were automatically a monochromatic union-closed family in the sense of this paper, the polynomial bound would immediately finish the first conjecture: taking k with $k^2 - k + 2 \leq n$, i.e. $k = \Theta(\sqrt{n})$, every colouring of $2^{[n]}$ would contain a monochromatic union-closed family of size $2^k = 2^{\Theta(\sqrt{n})}$, which exceeds n^c for every fixed c once n is large. This does not work, and the reason is exact rather than a matter of degree: an order-embedding need not be a lattice embedding. The constructions in [AW17, LT22] produce ϕ by a chain decomposition of $2^{[n]}$ into symmetric chains, walking up the chains level by level and, at each level, selecting an available element of the right colour *from whichever chain happens to offer one*; nothing in the argument arranges for $\phi(A) \cup \phi(B)$ to equal $\phi(A \cup B)$ for $A, B \subseteq [k]$, or even for $\phi(A) \cup \phi(B)$ to be monochromatic, let alone to lie in the image of ϕ at all. A copy of Q_k obtained this way witnesses 2^k elements of $2^{[n]}$ correctly ordered by inclusion and of one colour; it does not witness a union-closed family, since union-closure is a statement about actual unions in $2^{[n]}$, not about the abstract order type of a chosen set of representatives. Repairing this would require either forcing the chain-selection to respect a fixed block structure throughout, which the published proofs do not do and which does not obviously survive the level-by-level, chain-local nature of the argument, or a genuinely different construction; the authors were not able to find one, and are not aware of one in the literature. The gap between the two notions of “copy” is therefore, at present, exactly the gap between what is known about poset Ramsey numbers and what the first conjecture requires.

Remark 10.5 (A second approach that does not work, and precisely why). Unlike the previous approach, there is a construction that genuinely avoids the defect of Remark 10.4: pairwise disjoint nonempty $X_1, \dots, X_d \subseteq [n]$ generate, by $I \mapsto \bigcup_{i \in I} X_i$, an actual sublattice of $2^{[n]}$ isomorphic to $2^{[d]}$, a d -dimensional

Boolean algebra in the terminology of Gunderson, Rödl and Sidorenko [GRS99]; if this copy is monochromatic, its $2^d - 1$ nonempty members are a genuine monochromatic union-closed family (indeed a sublattice), with no gap between “copy” and “union-closed family” to repair. The relevant Ramsey function, $r(d, n)$, is the largest r such that every r -colouring of the subsets of an n -set contains a monochromatic d -dimensional Boolean algebra; Gunderson, Rödl and Sidorenko prove

$$c_1 n^{1/2^d} \leq r(d, n) \leq n^{d/2^{d-1}(1+o(1))}$$

for an absolute constant $c_1 > 0$ ([GRS99, Theorem 4.7]; the case $d = 2$ is their Theorem 4.6, with the sharper constants $(1 - o(1))\frac{3}{4}\sqrt{n} \leq r(2, n) \leq (1 + o(1))\sqrt{n}$). Fixing $r = 2$ and asking how large a d this forces at a given n is exactly the question needed for the first conjecture, and both bounds answer it the same way: the lower bound shows $r(d, n) \geq 2$, so that the guarantee holds, as soon as $n \geq (2/c_1)^{2^d}$, while the upper bound shows a 2-colouring avoiding every monochromatic d -dimensional Boolean algebra exists as soon as $n^{2d/2^d(1+o(1))} < 2$, i.e. for n below essentially the same threshold. Both directions therefore agree that the threshold is *doubly* exponential in d , $n = \Theta(2^{2^d})$ up to the constant in the exponent, not tower-type as in Remark 10.4’s more distant relative, the Hilbert cube problem [Gow01, Bal19], but also not singly exponential. Inverting, the largest d that a 2-colouring of $2^{[n]}$ is guaranteed to force by this method is only $d(n) = (1 + o(1)) \log_2 \log_2 n$, and the resulting monochromatic union-closed family has only $2^{d(n)} - 1 = \Theta(\log n)$ members: smaller, for every large n , than the trivial chain bound $\lceil (n + 1)/2 \rceil$ of (1.1). Because the upper bound in [GRS99] matches the lower bound in the same doubly-exponential shape, this is not a matter of a Ramsey number not yet being computed well enough: for a fixed pair of colours, monochromatic Boolean subalgebras generated by disjoint blocks are, provably, too weak a structure to beat a single chain, let alone to grow polynomially with n . Any resolution of the first conjecture must therefore use union-closed families that are not sublattices of this particularly symmetric, block-generated kind.

Both remarks above attack the first conjecture directly, over all of $2^{[n]}$ at once. The last attempt instead tries to build it up by recursion on n , and the recursion turns out to be exactly as hard as the problem it is meant to simplify – provably so, not just in practice.

LEMMA 10.6 (Doubling, conditional on a simultaneous family). *Let $n \geq 1$, let χ be a 2-colouring of $2^{[n]}$, and define $\chi_0, \chi_1 : 2^{[n-1]} \rightarrow \{0, 1\}$ by $\chi_0(A) =$*

$\chi(A)$ and $\chi_1(A) = \chi(A \cup \{n\})$. If there is a union-closed family $F \subseteq 2^{[n-1]}$ and a colour c such that every member of F has colour c under both χ_0 and χ_1 , then

$$G = F \cup \{A \cup \{n\} : A \in F\}$$

is a monochromatic (colour c) union-closed family for χ , and $|G| = 2|F|$.

Proof. G has colour c throughout, by hypothesis. For closure under union, take $P, Q \in G$. If $P, Q \in F$ then $P \cup Q \in F \subseteq G$ since F is union-closed. If $P = A \cup \{n\}$ and $Q = A' \cup \{n\}$ with $A, A' \in F$, then $P \cup Q = (A \cup A') \cup \{n\} \in G$ since $A \cup A' \in F$. If $P = B \in F$ and $Q = A \cup \{n\}$ with $A \in F$, then $P \cup Q = (A \cup B) \cup \{n\} \in G$ since $A \cup B \in F$. The two halves of G are disjoint, since one avoids n and the other contains it, so $|G| = 2|F|$. $\square$

If the lemma's hypothesis – a single union-closed family simultaneously monochromatic, in the same colour, under two given colourings of $2^{[n-1]}$ – could always be met with $|F| \geq F(n-1)$, then applying it at every level from some fixed n_0 up would give $F(n) \geq 2^{n-n_0} F(n_0)$ for all $n \geq n_0$, which is $2^{\Omega(n)}$. This contradicts Theorem 1.2: $F(n) \leq n^{(1+o(1)) \log_2 n} = 2^{(1+o(1))(\log_2 n)^2} = 2^{o(n)}$. Hence the required simultaneous family fails to exist, with $|F| \geq F(n-1)$, for infinitely many n – if it failed only finitely often, the recursion could still be run from the last failure onward and the same contradiction would follow. This is not an empirical difficulty with the three patches attempted below; it is a proof, from the paper's own Theorem 1.2, that no uniform way of supplying the lemma's hypothesis at every level can exist.

Remark 10.7 (A third approach that does not work, and precisely why). The obstruction identified above still leaves open whether the simultaneous family can be found *often enough* – say, at all n outside a sparse exceptional set – to salvage a weaker recursion; the authors tried three routes to supplying it and each one breaks union-closure instead of repairing it. First, apply the first conjecture's guarantee to χ_0 alone to get a union-closed $F_0 \subseteq 2^{[n-1]}$ of size $\geq F(n-1)$ in some colour c_0 , then 2-colour F_0 itself by χ_1 : by pigeonhole at least $|F_0|/2$ members of F_0 share a single χ_1 -colour c_1 , but this majority subfamily need not be union-closed, since the union of two of its members can fall on the χ_1 -minority side. Second, treat $\psi = (\chi_0, \chi_1)$ as a single 4-colouring of $2^{[n-1]}$ and ask for a union-closed family monochromatic under ψ : this only helps if the forced ψ -colour happens to be a diagonal value (c, c) rather than $(0, 1)$ or $(1, 0)$, and nothing forces the diagonal. Third, split $[n]$ into two halves instead of peeling one point, and look slice by slice: each slice χ_A (for A ranging

over subsets of the first half) admits its own union-closed family by induction on half the size, but combining these into one union-closed family over all of $[n]$ requires many independently-obtained structures to agree, which is the same difficulty that gives Remark 10.5 its doubly-exponential cost, not a way around it. The recursion of Lemma 10.6 is therefore not merely unproved at every step; by the argument above it is *provably* unavailable at infinitely many steps, and the three natural repairs available to a coloring-agnostic argument all either destroy closure or reduce to the difficulty already isolated in Remark 10.5.

11. Formal verification in Lean

The definitions of $f(n)$ and $F(n)$, the first assertion of Theorem 1.2, the resulting resolution of Erdős’s second conjecture, and the trivial lower bound (1.1) have been formalised in the Lean 4 proof assistant [MU21] on top of the Mathlib library [Mat20]. The formalisation is the directory `lean/` of the repository accompanying the paper (<https://github.com/creelie/erdos-1183>); it builds with Lean 4.34.1 and Mathlib v4.34.1, contains no `sorry` and declares no axiom, and for every theorem listed below Lean’s `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

In the formalisation $[n]$ is `Fin n`, a colouring is a function `Finset (Fin n) → Bool`, and `bigF n` and `smallF n` are defined literally as in Section 1, as the minimum over colourings of the largest monochromatic family closed under union (respectively under union and intersection). The lemmas `le_bigF_iff` and `bigF_le_iff` check that `bigF n` is exactly the largest m such that every colouring contains a monochromatic union-closed family of m sets.

Lean theorem	Statement
<code>bigF_le_sum_choose</code>	if $nd + 2 < 2^d$ then $F(n) \leq \sum_{j < d} \binom{n}{j}$ (Theorem 1.2, first assertion)
<code>bigF_le_explicit</code>	for $n \geq 2$ and $d = 2(\lfloor \log_2 n \rfloor + 1)$, $F(n) \leq dn^d$
<code>second_conjecture</code>	for every real $\varepsilon > 0$, $F(n) < (1 + \varepsilon)^n$ for all sufficiently large n
<code>half_le_smallF</code> , <code>smallF_le_bigF</code>	$\lceil (n + 1)/2 \rceil \leq f(n) \leq F(n)$, i.e. (1.1)

Table 3. Machine-checked statements.

The formal proof follows Section 4 with two changes. The first-moment argument is replaced by the equivalent count: for a fixed independent d -tuple the $2^d - 1$ unions are distinct (Lemma 2.5(a)), so at most $2 \cdot 2^{2^n - (2^d - 1)}$ colourings make the tuple bad, and summing over the at most 2^{nd} tuples gives fewer than 2^{2^n} colourings when $nd + 2 < 2^d$. The Sauer–Shelah lemma is Mathlib’s (`Finset.card_le_card_shatterer` and `Finset.card_shatterer_le_sum_vcdim`). For the asymptotic statement the formalisation uses the simpler choice $d = 2(\lfloor \log_2 n \rfloor + 1)$, which gives $F(n) \leq n^{O(\log n)}$ and suffices for $F(n) < (1 + \varepsilon)^n$; the sharper exponent $(1 + o(1)) \log_2 n$ of Theorem 1.2 is not formalised.

Theorems 1.1, 1.3, 1.4 and 7.2 are not formalised; Theorem 1.4 rests on the satisfiability computations of Section 8. Erdős’s first conjecture is stated in the formalisation as the proposition `FirstConjecture` ($F(n) > n^c$ for every fixed c and all large n) and is not proved: it remains open.

Appendix A. The satisfiability encoding

This appendix formalises the algorithm sketched in Section 8, for a reader who wants to reproduce or re-implement it without reading the code.

A.1. *Variables and the colouring clauses.* Fix n . The Boolean variables are x_S for $S \subseteq [n]$, one per subset, with $x_S = 1$ meaning “ S has colour 1”. The symmetry $\chi \mapsto 1 - \chi$ is broken once and for all by the unit clause $\neg x_\emptyset$. No other clause is imposed at the outset; clauses are added only as counterexamples are found, in the loop described next.

A.2. *The closed-family clause.* For a family $\mathcal{F} \subseteq 2^{[n]}$ with $|\mathcal{F}| = M \geq 2$, forbidding $\mathcal{F}$ from being monochromatic is the pair of clauses

$$\left(\bigvee_{S \in \mathcal{F}} \neg x_S \right) \quad \text{and} \quad \left(\bigvee_{S \in \mathcal{F}} x_S \right),$$

the first true unless every $S \in \mathcal{F}$ has $x_S = 1$, the second true unless every $S \in \mathcal{F}$ has $x_S = 0$; together they say that $\mathcal{F}$ is not entirely one colour. There are $2^{2^n} - 2$ closed families of size ≥ 2 in the worst case (Lemma 2.3 bounds this far more sharply), far too many to write down in advance for $n \geq 5$ or so; the counterexample-guided loop adds only the ones that are actually needed.

A.3. *The outer loop.* For a target M , the loop maintains a growing CNF formula Φ , initialised to $\{\neg x_\emptyset\}$.

- (1) Call the SAT solver on Φ . If unsatisfiable, report that every colouring has a monochromatic closed family of size $\geq M$ (so the value being computed is $\geq M$) and stop.
- (2) Otherwise let χ be the returned model. For each colour $c \in \{0, 1\}$, run the inner search of Section A.4 below on the colour class $\chi^{-1}(c)$ with target M .
- (3) If neither inner search finds a closed family of size $\geq M$, χ is a witness that the value is $< M$; report χ and stop.
- (4) Otherwise, for each colour class in which the inner search found a closed family $\mathcal{F}$ of size $\geq M$, add the two clauses of Section A.5 for that $\mathcal{F}$ (or, if $|\mathcal{F}| > M$, for an M -element closed subfamily of it, which exists since removing a maximal element of a closed family under either operation leaves a smaller closed family) to Φ , and go to (1).

Termination follows because each pass through the loop either stops or adds at least one clause pair ruling out at least one previously satisfying model, and there are only 2^{2^n} models in total; in practice the loop terminates after examining a number of families in the tens or low hundreds of thousands (Section 8 records the exact counts for the two largest cases, $f(9)$ and $F(7)$), far fewer than 2^{2^n} .

A.4. *The inner search.* Given a set $C \subseteq 2^{[n]}$ (a colour class) and a target M , the inner search looks for a subfamily $\mathcal{F} \subseteq C$ with $|\mathcal{F}| \geq M$ that is closed under the relevant operations (union alone for F , union and intersection for f). It performs a depth-first branch-and-bound search that maintains a partial closed family, at each step either adding an unused element of C and closing (repeatedly adjoining unions, and for f also intersections, of existing members that lie in C , until no new member of C is produced) or backtracking; a branch is pruned as soon as the sum of the current family's size and the number of remaining candidates in C falls below M . This is exactly the sense in which $\mathcal{F}$ is “built by adding sets one at a time and closing under the operations” in Section 8.

A.5. *The independent verifier.* The witness-checking program of Section 8 re-expresses, for a fixed colour class C and target M , the existence of a closed subfamily of size $\geq M$ as a second, differently structured satisfiability instance: one Boolean variable y_S for each $S \in C$, the closure conditions $\neg y_S \vee \neg y_{S'} \vee y_{S \cup S'}$ (and, for f , the analogous clause for $S \cap S'$) for all $S, S' \in C$, and a cardinality constraint $\sum_{S \in C} y_S \geq M$ encoded by the sequential counter method of Sinz

[Si05], which introduces $O(|C|M)$ auxiliary variables and clauses rather than the $\binom{|C|}{M}$ clauses of a naive encoding. When $S \cup S' \notin C$ (respectively $S \cap S' \notin C$), the corresponding closure clause degenerates to $\neg y_S \vee \neg y_{S'}$, since no third variable is available to witness closure and S, S' simply cannot be chosen together; this is the same convention used, without comment, in the code. This instance shares no variables, clauses, or code with the branch-and-bound search of Section A.4; agreement between the two, checked for every entry of Theorem 1.4 (Section 8), is what makes the values independently verified rather than merely computed once.

Competing Interests

The authors declare that there are no competing interests, financial or non-financial, that could have influenced the research reported in this paper.

Data Availability

The program computing the values in Theorem 1.4, with its output, and the Lean formalisation of Section 11 accompany the paper and are available at <https://github.com/creelie/erdos-1183>. The underlying data and code are also available from the corresponding authors upon reasonable request by email.

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FORMERLY, ELECTRO-GRAVITATIONAL SPACE PROPULSION LABORATORY (EGSPL),
BHUBANESWAR, ODISHA 751030, INDIA

E-mail: d.bhattacharjee@erl-forschung.de

E-mail: itsdeep@live.com

ORCID: [0000-0003-0466-750X](https://orcid.org/0000-0003-0466-750X)

DEPARTMENT OF MATHEMATICS, BIOINFORMATICS AND COMPUTER APPLICATIONS,
MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL, BHOPAL 462003, INDIA

E-mail: priyabrata@manit.ac.in

ORCID: [0000-0001-6472-6239](https://orcid.org/0000-0001-6472-6239)

FORMERLY, NATIONAL TAIWAN UNIVERSITY, TAIPEI 106319, TAIWAN

E-mail: rai.bhattacharya.in@gmail.com

ORCID: [0009-0002-2254-3914](https://orcid.org/0009-0002-2254-3914)